



Calculator

Examination

General
introduction

Combinatorics

Lecture notes for Applies statistics, Probability theory

Lecture 1.

Dr. Barta Attila¹, Dr. Pecsora Sándor²

^{1,2}University of Debrecen, Faculty of informatics,
Department of Applied statistics and probability theory

CS BSc, CSE BSc



Presentation Overview

Calculator

Examination

General
introduction

Combinatorics

Calculator

Examination

General introduction

Combinatorics



Calculator

Calculator

Examination

General
introduction

Combinatorics

No, You can not use your or any calculators!



Examination

Calculator

Examination

General
introduction

Combinatorics

All the important information about the examination, laboratory classes, lectures, attendance, and general behaviour can be found in the syllabus of the course.



General introduction

Calculator

Examination

General
introduction

Combinatorics

Probability Theory *or* Statistics

Probability Theory *and* Statistics

Probability Theory *V/S* Statistics

Probability theory deals with predicting the likelihood of **future events**, while *statistics* involves the analysis of the frequency of *past events*. In practice statistics built on the foundations created by the results of probability theory.



Concept Question

8 students registered for the Probability theory exam. How many different orders can be made?

- ☐ 64
- ☐ 16
- ☐ 40320
- ☐ 34862

☐☐☐☐☐☐☐☐



Concept Question

8 students registered for the Probability theory exam. How many different orders can be made?

- ☐ 64
- ☐ 16
- ☒ 40320
- ☐ 34862

8	7	6	5	4	3	2	1



Concept Question

DNA is made of sequences of nucleotides: A, C, G, T. How many DNA sequences of length 3 are there?

- ☐ 12
- ☐ 24
- ☐ 64
- ☐ 81





Concept Question

DNA is made of sequences of nucleotides: A, C, G, T. How many DNA sequences of length 3 are there?

- ☐ 12
- ☐ 24
- ☒ 64
- ☐ 81

$4 \times 4 \times 4$



Concept Question

On the state exam there are 20 questions. How many different selections can be made, if the students have to choose 2 questions?

- 190
- 256
- 39
- 380





Concept Question

On the state exam there are 20 questions. How many different selections can be made, if the students have to choose 2 questions?

- **190**
- 256
- 39
- 380

$\frac{20}{2} = 19$
☐ ☐

Does the order matter???



Definition (Permutation)

An ordered sequence of n distinguishable objects is called an n -permutation.



Definition (Permutation)

An ordered sequence of n distinguishable objects is called an n -permutation.

Theorem

The number of n -permutations is n -factorial, that is

$$P_n = n! = n \cdot (n-1) \cdots 2 \cdot 1$$

Remark:

$$0! = 1$$

Since the number of possible orders of the elements of an empty set is 1.



Definition (Permutation with repetition)

An ordered sequence of n elements, where we have m different types of elements, such as the first type has k_1 elements, the second type has k_2 elements, $\dots$, the m^{th} type got k_m elements is called an n permutation with $k_1, k_2, \dots, k_m$ repetitions.



Definition (Permutation with repetition)

An ordered sequence of n elements, where we have m different types of elements, such as the first type has k_1 elements, the second type has k_2 elements, $\dots$, the m^{th} type got k_m elements is called an n permutation with $k_1, k_2, \dots, k_m$ repetitions.

Theorem

The number of n -permutations with $k_1, k_2, \dots, k_m$ repetitions is

$$P_n^{k_1, k_2, \dots, k_m} = \frac{n!}{k_1! \cdot k_2! \cdot \dots \cdot k_m!}$$



Concept Question

Find the number of possible arrangements of 8 castles on the chess board in a way that they do not hit each other?



Concept Question

Find the number of possible arrangements of 8 castles on the chess board in a way that they do not hit each other?

$$P_8 = 8!$$



Concept Question

Find the number of possible arrangements of 8 castles on the chess board in a way that they do not hit each other?

$$P_8 = 8!$$

What is the result if we can distinguish between the castles?



Concept Question

Find the number of possible arrangements of 8 castles on the chess board in a way that they do not hit each other?

$$P_8 = 8!$$

What is the result if we can distinguish between the castles?

$$P_8 \cdot P_8$$



Definition (Ordered selection)

Ordered selections without replacement. Choose k out of n distinguishable objects and order them.



Definition (Ordered selection)

Ordered selections without replacement. Choose k out of n distinguishable objects and order them.

Theorem

The number of ordered selections without replacement is

$$V_n^k = n \cdot (n-1) \cdot (n-2) \cdots k = \frac{n!}{(n-k)!}$$



Ordered selection with repetition

Calculator

Examination

General
introduction

Combinatorics

Definition (Ordered selection with repetition)

Choose k out of n distinguishable elements so that each and every element can be chosen multiple times. The order of the elements does matter.



Ordered selection with repetition

Calculator

Examination

General
introduction

Combinatorics

Definition (Ordered selection with repetition)

Choose k out of n distinguishable elements so that each and every element can be chosen multiple times. The order of the elements does matter.

Theorem

The number of ordered selections with replacement is

$$V_n^{k,r} = \underbrace{n \cdot n \cdots n}_{k \text{ times}} = n^k$$



Concept Question

How many real four digit numbers (they can not start with zero) can be formed from digits 0, 1, 2, 3, 4, 5, 6?



Concept Question

How many real four digit numbers (they can not start with zero) can be formed from digits 0, 1, 2, 3, 4, 5, 6?

$$V_6^1 \cdot V_6^3$$



Concept Question

Three postmen must deliver six letters. Find the number of possible distribution of the letters.



Concept Question

Three postmen must deliver six letters. Find the number of possible distribution of the letters.

$$V_3^{6,r}$$



Definition (Combination without repetition)

Combination is a selection of items from a set, that has distinguishable elements and the order of selection does not matter.



Definition (Combination without repetition)

Combination is a selection of items from a set, that has distinguishable elements and the order of selection does not matter.

Theorem

The number of ways to choose k objects out of n distinguishable objects is

$$C_n^k = \binom{n}{k}$$

Here n choose k or the binomial coefficient is

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$



Combination with repetition

Calculator

Examination

General
introduction

Combinatorics

Definition (Combination with repetition)

Let us have n distinguishable elements, we choose k elements so that each and every element can be chosen multiple times.



Definition (Combination with repetition)

Let us have n distinguishable elements, we choose k elements so that each and every element can be chosen multiple times.

Theorem

The number of ways to choose k elements out of n distinguishable elements so that each and every element can be chosen multiple times is

$$C_n^{k,r} = \binom{n+k-1}{k}$$



Concept Question

Find the number of possible fillings of a lottery coupon (5 numbers from 90)!



Concept Question

Find the number of possible fillings of a lottery coupon (5 numbers from 90)!

$$C_{90}^5 = \frac{90!}{5! \cdot (90 - 5)!}$$



Concept Question

In how many ways we can distribute 7 apples and 9 peaches among 4 kids?



Concept Question

In how many ways we can distribute 7 apples and 9 peaches among 4 kids?

$$C_4^{7,r} \cdot C_4^{9,r}$$